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A LEARNING RESOURCE RECOMMENDATION ALGORITHM BASED ON ONLINE LEARNING BEHAVIOR

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ABSTRACT

Faced with abundant online course resources, learners struggle to choose suitable materials. Learning resource recommendation algorithms can help address this. Rich online learning behavior data enables such recommendations. However, existing research only uses behavior events as learner features, ignoring event order i.e. Learning Behavior Patterns (LBPs). Also, only using click counts loses valuable information, hurting performance. We propose an algorithm leveraging online behavior sequences. First, extract sequences from logs and generate LBPs. Next, calculate Term Frequency Inverse Document Frequency (TF-IDF) values for each LBP as feature vectors. Cluster learners to improve efficiency. Finally, Calculate intra-cluster similarities for collaborative filtering recommendations. Experiments show over 30% precision, 9% recall, and 10% F1 improvements versus existing methods. Further ablation indicates learner clustering boosts time efficiency 3.75x without performance impact. Using TF-IDF values and tuning LBP length significantly improves performance. Overall, modeling orders via LBPs and better features like TF-IDF give major gains.

Index Terms— Learning resource recommendation, Online Learning Behavior, Learning Behavior Patterns, User Clustering, Collaborative Filtering

1. INTRODUCTION

Online learning systems provide new methods for acquiring knowledge and have been expanding their popularity and influence in recent decades. The continuous enrichment of learning resources and the accessibility of massive learning resources have also caused problems such as ”information overload”, ”knowledge maze”, and ”learning topic drift”[1]. Without proper guidance, learners may encounter difficulties in selecting appropriate learning resources in the face of a large amount of information in the learning process[2], and the Learning Resource Recommendation (LRR) algorithm provides a solution to this problem[3].

The rich system logs in the online learning system make it possible to recommend more detailed granularity learning resources. These system logs record the learning behavior events of learners in the learning system[4], such as viewing documents, conducting tests, and playing videos. Many studies use online behavior data to predict learning styles[5], and students at risk[6], but few studies involve LRR. For example, Chen et al.[7] use learning behavior data to characterize learners' learning styles, use the number of clicks of learning behavior events as the feature value, then divide learners into different clusters through clustering algorithms, and use association rules and collaborative filtering algorithms to recommend learning resources for learners. Some researchers also use learners' online learning sequential behavior as their features and combine them with clustering algorithms to recommend resources for learners[8]. Additionally, studies have analyzed students' ability levels based on the number of times they pause educational videos, the playback speed of these videos, and the frequency of fast-forwarding within the videos to recommend subsequent learning videos[9]. Research has also been conducted utilizing metrics such as the number of clicks on learning resources, duration of tasks, frequency of text queries and re-queries, and user feedback ratings to improve recommendations[10].

However, current LRR algorithms based on online learning behavior take the number of clicks on learning resources as the feature value when characterizing student features[11]. There are two disadvantages to this approach. Firstly, the chronological order of different learning behavior events is not considered, and the relationship between behavior events is ignored. Therefore, this study proposes to use Learning Behavior Patterns (LBPs) as the characteristics of learners[12]. LBPs refer to the sequence of learning behavior events, which represents the behavior patterns of learners in the learning process, and can more effectively characterize learners[13]. Secondly, simply treating the number of clicks as a feature value can result in less frequent but more important behavior patterns not being accurately characterized, making it difficult to achieve improved model performance. Therefore, this study draws inspiration from the commonly used weighting

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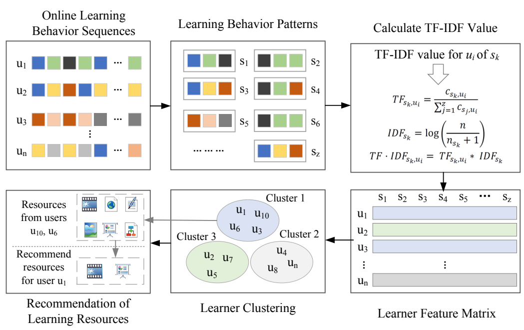

Fig. 1. The Architecture of the Proposed Method.

technique TF-IDF algorithm in information retrieval and data mining[14], which uses LBPs as words and learners' learning behavior sequences as documents to calculate the TF-IDF values for each LBP.

In this paper, we propose a Learning Resource Recommendation Algorithm based on Online Learning Behavior (LRRA-OLB). Firstly, the algorithm extracts the learning behavior sequence of each learner from the system logs. Second, extract the LBPs from the learning behavior sequence of learners. Third, calculate the TF-IDF values of all LBPs of all learners and construct the learner feature matrix through our proposed TF-IDF value calculation method for LBPs. Fourthly, in order to improve time efficiency, clustering algorithms are used to divide learners into K different clusters and identify M learners in the same cluster that are most similar to the target learner. Finally, based on the collaborative filtering algorithm, N learning resources are recommended for the target learner. The source code used in the paper is available at https://github.com/xingyezn/LRRA-OLB. The main contributions of our work are: 1) Propose a new learning resource recommendation algorithm that leverages learning behavioral data, demonstrating superior performance compared to existing baseline models. 2) Utilize learning behavior patterns as features for learners, and experimental results indicate that using LBPs is more effective in characterizing learners. 3) Introduce a TF-IDF calculation method for the LBPs features of learners, and results show that using TF-IDF leads to better outcomes.

2. METHOD

2.1. Extracting Learning Behavior Sequences

Our proposed LRRA-OLB method is shown in Fig. 1. Users studying on online learning platforms are defined as learners, and one learner is written as u_i, $0 < i \leq n$, n indicates the total number of learners. The set of learning behavior events is defined as $E = \{e_1, e_2, ..., e_q\}$, q is the number of learning behavior event types. We extract the online learning behavior sequence of each learner u_i from the system logs and organize these behavior events in chronological order to form the learner's online learning behavior sequence $B_i = \{b_{i,1}, b_{i,2}, ..., b_{i,p}\}$, $b_{i,p} \in E$, which represents a learning behavior event, $0 < p \leq h$, h is the total number of learning behavior events of u_i.

2.2. Extracting Learning Behavior Patterns

Studies show online behavior reflects learning preferences[7]. LBPs are repeated behavior patterns in the sequence of learners' behavior. These patterns can represent learners' characteristics in more detail. The acquisition of LBPs can be constructed based on theoretical research, prior knowledge, or data-driven approaches. In this study, we traverse the online learning behavior sequence of all learners, set the window length to L to slice the behavior sequence, and then obtain the set $S = \{s_1, s_2, ..., s_k\}$, which represents all LBPs found from the behavior sequences of all learners. A s_k refers to a pattern of learning behavior events that repeatedly occur in the behavior sequences of learners. Where $0 < k \leq z$, z represents the total number of LBPs, which also is the number of learner features.

2.3. Generate Learner Feature Matrix

By obtaining all LBPs, we can construct the learner feature matrix. The current approaches take the number of occurrences of LBPs as feature values, but this method cannot accurately represent their importance. For example, a learner has two LBPs: $s_1 = \{e_1, e_3, e_4\}$ and $s_2 = \{e_1, e_2, e_3\}$. There are 100 occurrences of s_1 and 20 occurrences of s_2. Although s_1 seems more important, in fact, 99% of learners have s_1, while only 10% have s_2. Obviously, s_2 can better reflect the characteristics of the learner. We use the idea of the TF-IDF algorithm commonly used in the field of natural language processing for reference and build a TF-IDF calculation method for LBPs. Here, we treat a LBP s_k as a word, a learner's behavior sequence B_i as a document, and all learners' behavior sequences set B as a corpus, while set S is the set of all words in the corpus. The formula for calculating the TF-IDF value t_i, k of the LBP s_k corresponding to learner u_i is as follows:

$$TF_{s_k,u_i} = \frac{c_{s_k,u_i}}{\sum_{j=1}^{z} c_{s_j,u_i}} \tag{1}$$

$$IDF_{s_k} = \log\left(\frac{n}{n_{s_k}+1}\right) \tag{2}$$

$$TF - IDF_{s_k,u_i} = TF_{s_k,u_i} * IDF_{s_k} \tag{3}$$

In (1), c_{s_j,u_i} represents the number of occurrences of LBP s_j in the online behavior sequence B_i of learner u_i, where $0 < j \leq z$. In (2), n is the total number of learners, and n_{s_k} is the number of learners with LBP s_k. Observing (3), we can know that if s_k does not appear in B_i, its corresponding $TF_{s_k,u_i} = 0$, therefore its TF-IDF value is also 0. In our formula, TF reflects the importance of LBPs to the learner, while IDF represents the uniqueness of LBPs among learners. Through the above method, we can calculate the feature vector x_i of learner $x_i = \{t_{i,1}, t_{i,2}, ..., t_{i,z}\}$, and obtain the feature matrix $X = \{x_1, x_2, ..., x_i\}$ for all learners.

2.4. Learner Clustering

For learner clustering, a common method involves initially assigning learners to various groups. When calculating the similarity between the target learner and others, only the relevant groups are considered, eliminating the need to traverse all learners and significantly improving time efficiency[15]. Given a set of n learners' feature matrix $X = \{x_1, x_2, ..., x_i\}$, we use Euclidean metric to calculate the distance between two learners. By measuring the distance, K-means algorithm can divide n learners into K disjoint clusters $C = \{c_1, c_2, ..., c_k\}$, each cluster c_k is a set of learners. The main idea of K-means is to update the centroids by iterative computation until some criteria for convergence are met.

2.5. Recommendation Based on Similar Learners

After learner clustering, we recommend learning resources for target learners based on similar learners. For a given target learner u_i, we calculate the similarity between u_i and other learners in its cluster c_k, and select the top M learners with the highest similarity as the set of similar learners $T_{u_i} = \{u_1, u_2, ..., u_m\}$, $u_M \in c_k$. The $sim(u_m, u_n)$ represents the Pearson similarity between two learners u_m and u_n by (5), where the feature vectors of the two learners are $x_m = \{t_{m,1}, t_{m,2}, ..., t_{m,z}\}$ and $x_n = \{t_{n,1}, t_{n,2}, ..., t_{n,z}\}$, $\bar{x_m}$ represents the mean of x_m, $\bar{x_n}$ represents the mean of x_n. The set of learning resources is defined as $R = \{r_1, r_2, ..., r_a\}$, where r_a represents a learning resource that can be a teaching video, PDF document, or presentation, and a represents the total number of learning resources.

$$sum\left(u_m, u_n\right) = \sum_{i=1}^{z}\left(t_{m,i} - \bar{x_m}\right)^2 * \sum_{i=1}^{z}\left(t_{n,i} - \bar{x_n}\right)^2 \quad (4)$$

$$sim\left(u_m, u_n\right) = \frac{\sum_{i=1}^{z}\left(t_{m,i} - \bar{x_m}\right) * \left(t_{n,i} - \bar{x_n}\right)}{\sqrt{sum\left(u_m, u_n\right)}} \quad (5)$$

$$I\left(r_a, u_i\right) = \sum_{m=1}^{M} sim\left(u_i, u_m | r_a\right), u_m \in T_{u_i} \quad (6)$$

Then, we define the degree of interest of learner u_i on resource r_a by (6), $sim\left(u_i, u_m | r_a\right)$ represents the similarity between the learner u_m who has clicked on the learning resource r_a and the target learner u_i. By calculating the interest of u_i in all learning resources and removing the learning resources that u_i has clicked on, the top N learning resources with the highest interest are selected and recommended to u_i.

3. EXPERIMENT

3.1. Datasets and Evaluation Metric

To make results more reliable, an open-source dataset is chosen for evaluation. Open University Learning Analysis Dataset[16] is an open-source dataset that contains information about 22 courses, 32,593 students. The dataset has the logs of students' interactions with the virtual learning environment represented by daily summaries of student clicks (10,655,280 entries) from 2013 and 2014, including twenty kinds of resource types are recorded. We use Precision, Recall, and F1 scores to measure the performance of the algorithm, which are widely used in LRR to evaluate the quality of recommendations[17].

3.2. Experimental Setup

In this experiment, the performance of LRRA-OLB is compared with that of the baseline methods, including the user-based CF method (UserCF), the item-based CF method (ItemCF), the AROLS method[7] and the PLRec-KL method [11]. Evaluate the performance of all methods using the above evaluation metrics. In LRRA-OLB, the length of the LBP is $L = 1$, the number of similar learners $M = 40$, and the number of clusters $K = 4$. We use Silhouette Coefficient (SC) [18] and Calinski Harabaz (CH) [19] scores to measure the quality of clustering results and determine the number of clusters K. The SC score is bounded between 1 for incorrect clustering and -1 for highly dense clustering. The high CH score relates to a model with well-defined clusters. When $K = 4$, $SC = 0.406$, and $CH = 9091.057$ scores are both maximum points, we believe that the best clustering result is achieved when $K = 4$. To further analyze our method, we conducted ablation experiments to explore the performance effects of different operations such as not using clustering, not using TF-IDF values, and LBPs of different lengths.

3.3. Comparison of Different Methods

The results of different methods are shown in Table 1. It can be found that the precision value decreases as N increases, while recall and F1 are the opposite. 1) Compared with the baseline method, LRRA-OLB achieved the highest precision on all N. For example, when N is equal to fifteen, the highest baseline method, PLRec-KL, has a precision of only 0.309, while our method is twice that of 0.619. It indicates that LRRA-OLB can significantly improve the precision of LRR algorithms, which is crucial for LRR tasks, as learners always expect to be recommended the learning resources they are interested in and need. 2) The recall of the LRRA-OLB is

Table 1. The Performances of Different Methods

N	ItemCF	UserCF	AROLS	PLRec-KL	LRRA-OLB
			Precision		
5	0.187	0.126	0.281	0.303	**0.644**
15	0.173	0.127	0.279	0.309	**0.623**
25	0.158	0.118	0.269	0.294	**0.601**
35	0.145	0.109	0.258	0.279	**0.579**
45	0.136	0.097	0.246	0.275	**0.558**
55	0.132	0.090	0.244	0.267	**0.536**
			Recall		
5	0.042	0.057	0.069	**0.131**	0.054
15	0.045	0.061	0.070	0.137	**0.150**
25	0.051	0.065	0.073	0.142	**0.232**
35	0.054	0.070	0.075	0.145	**0.304**
45	0.056	0.072	0.076	0.148	**0.367**
55	0.061	0.072	0.075	0.151	**0.422**
			F1 Score		
5	0.069	0.078	0.111	**0.183**	0.094
15	0.071	0.082	0.112	0.190	**0.220**
25	0.077	0.084	0.115	0.192	**0.301**
35	0.079	0.085	0.116	0.191	**0.356**
45	0.079	0.083	0.116	0.192	**0.395**
55	0.083	0.080	0.114	0.193	**0.421**

higher than other baseline methods when N is greater than five, indicating that the LRRA-OLB exhibits better recall performance in most cases, that is, the proportion of correctly recommended resources to the number of learning resources that learners have clicked on is higher. When N is equal to five, due to the small number of recommended learning resources, the proportion of correctly recommended resources in the total learning resources of learners is relatively small. 3) The F1 score of the LRRA-OLB method is higher than other baseline methods when N is greater than five, and only lower than the AROLS and PLRec-KL methods when N is equal to five, because the recall value of the LRRA-OLB method is lower at this time, which leads to a lower F1 score.

3.4. Ablation Experiment

The results of ablation experiment As shown in Fig. 2. 1) The impact of using clustering on algorithm performance was compared. Comparing the precision, recall, and F1 curves of LRRA-OLB and LRRA-OLB NO CLUSTER, it can be found that the two are almost completely coincident, indicating that using clustering does not reduce the performance of the algorithm. At the same time, when calculating the similarity between 19294 learners, using clustering took 5.16 seconds, while not using clustering took 19.34 seconds, which increased the time efficiency by 3.75 times. 2) It can be found that LRRA-OLB is significantly higher than LRRA-OLB NO TF-IDF in all three metrics. This also indicates that using TF-IDF values instead of clicks is effective, as it not only considers the importance of the LBPs for each learner, but also considers the importance of the LBPs for the overall learner, which can more effectively represent learner features. 3) To investigate the impact of different LBP lengths on algorithm performance, we used a private dataset from an online learning platform. This dataset collected 11,130,277 interactive records of twenty-one learning behavior events sorted by time from 4595 students in four courses, covering a period of one month. When the parameters $L \in [1, 7]$, $N = 40$, $k = 4$, we can see that increasing the length of the LBP can continuously improve the precision of the algorithm, as well as slightly improve the recall and F1 score of the algorithm. This shows that LBPs, as learner features, can improve the performance of the algorithm, because these LBPs can better represent the learning preferences of learners.

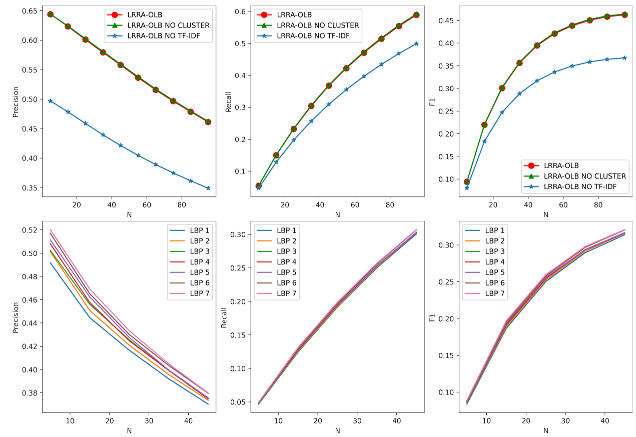

Fig. 2. The Results of Ablation Experiment

4. CONCLUSION

This article proposes a LRR algorithm based on online learning behavior. Through experiments, it is found that 1) our proposed algorithm outperforms existing baseline algorithms in various performance aspects. 2) Calculating the similarity of learners in different clusters through learner clustering can greatly improve the temporal performance of the algorithm without affecting its performance. 3) Using TF-IDF values as learner feature values can significantly improve the performance of the algorithm. 4) Choosing a LBP of appropriate length can also improve the accuracy of the algorithm. Finally, we believe that the method proposed in this article is a relatively universal method for extracting learner features, which can be used for both LRR and student performance prediction or students at risk prediction[6]. In addition, existing studies have shown that using user feedback data can further improve the performance of algorithms[20], which is also one of the future research directions.

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